🚀 Deployment Guide - Streamlit Cloud

This guide will help you deploy your User Dashboard to Streamlit Cloud in minutes.

✅ Pre-Deployment Checklist

Before deploying, make sure you have:

- [ ] All code files in your project directory
- [ ] A GitHub account
- [ ] Your Gemini API key
- [ ] Tested the app locally (`streamlit run app.py`)

📋 Step-by-Step Instructions

Step 1: Prepare Your Repository

1. **Create a GitHub repository**
   - Go to github.com
   - Click "New repository"
   - Name it: `ai-feedback-user-dashboard` (or any name)
   - Keep it Public (required for free Streamlit hosting)
   - Don't initialize with README (we have one)
2. **Push your code to GitHub**

```bash
```

```
# Initialize git (if not already done)
git init

# Add all files
git add .

# Make first commit
git commit -m "Initial commit - AI Feedback User Dashboard"

# Add remote repository
git remote add origin https://github.com/YOUR_USERNAME/YOUR_REPO_NAME.git

# Push to GitHub
git branch -M main
git push -u origin main
```

Important: Make sure `.env` is in `.gitignore` so your API key is NOT uploaded!

Step 2: Deploy to Streamlit Cloud

1. **Go to Streamlit Cloud**
   - Visit: share.streamlit.io
   - Click "Sign in" → Sign in with GitHub
   - Authorize Streamlit to access your repositories
2. **Create New App**
   - Click "New app" button
   - Select your repository from the dropdown
   - Choose branch: `main`
   - Main file path: `app.py`
   - (Optional) Custom URL: Choose a subdomain
3. **Advanced Settings (IMPORTANT)**
   - Click "Advanced settings"
   - Python version: `3.9` or higher
   - Add these dependencies are auto-detected from `requirements.txt`

Step 3: Add Your API Keys & Database Credentials (Secrets)

⚠️ **CRITICAL STEP -** Your app won't work without this!

1. **In Streamlit Cloud Dashboard**
   - After creating the app, click on your app
   - Click "⚙️ Settings" → "Secrets"
2. **Add Your Secrets**
   - Copy-paste this format:

```toml
GEMINI_API_KEY = "your_actual_gemini_api_key_here"
SUPABASE_URL = "https://your-project-id.supabase.co"
SUPABASE_KEY = "your_supabase_anon_key_here"
```

- Replace with your actual keys:
  - Gemini: From Google AI Studio
  - Supabase: From your Supabase project settings
- Click "Save"

3. **Verify**
   - Your secrets should look like this:

```
GEMINI_API_KEY = "AIzaSyB..."
SUPABASE_URL = "https://abc123.supabase.co"
SUPABASE_KEY = "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
```

Note: The app will work with just `GEMINI_API_KEY` (using CSV), but Supabase is recommended for production!

Step 4: Deploy!

1. **Click "Deploy"**
   - Streamlit will start building your app

- This takes 2-5 minutes on first deploy

2. **Watch the Build Logs**
   - You'll see:
     - Installing dependencies
     - Starting the app
     - App running at URL
3. **Your App is Live!** 🎉
   - You'll get a URL like: `https://your-app-name.streamlit.app`
   - Share this URL for your assignment submission

🔧 Post-Deployment

Testing Your Deployed App

1. Visit your Streamlit app URL
2. Try submitting a test review:
   - Rating: 5 stars
   - Review: "Amazing food and service!"
3. Verify:
   - ✅ AI response appears
   - ✅ Feedback is saved
   - ✅ Recent feedback shows up

Common Issues & Fixes

Issue 1: "GEMINI_API_KEY not found" or "SUPABASE_URL not found"

Solution:

- Go to Settings → Secrets
- Add missing keys in correct format
- Restart the app

Issue 2: "ModuleNotFoundError"

Solution:

- Check `requirements.txt` has all dependencies
- Push update to GitHub
- Streamlit will auto-redeploy

Issue 3: App keeps restarting

Solution:

- Check the logs in Streamlit dashboard
- Usually an error in code
- Fix locally, push to GitHub

Issue 4: Data not persisting between restarts

Solution:

- **With Supabase**: Data persists automatically ✅
- **Without Supabase**: CSV resets on Streamlit Cloud restart (expected)
- **Recommendation**: Use Supabase for production deployments

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🔄 Updating Your Deployed App

Streamlit Cloud automatically redeploys when you push to GitHub!

```bash
```

```
# Make changes to your code
# Test locally
streamlit run app.py

# Commit and push
git add .
git commit -m "Updated feature X"
git push

# Streamlit will auto-redeploy in ~1-2 minutes!
```

📊 Monitoring Your App

View Analytics

- Streamlit dashboard shows:
  - App uptime
  - Number of visitors
  - Resource usage

View Logs

- Click "Manage app" → "Logs"
- See real-time Python output
- Debug issues

🎯 Final Submission Checklist

For your assignment submission, include:

- [ ] **GitHub Repository URL**
  - Example: `https://github.com/username/ai-feedback-dashboard`
- [ ] **Deployed App URL**
  - Example: `https://ai-feedback-dashboard.streamlit.app`
- [ ] **Screenshots** (for report):

- Homepage with rating selector
- Review submission
- AI response display
- Recent feedback section

☐ **Test the app** one final time before submitting

🔐 Security Best Practices

✅ DO:

- Use Streamlit Secrets for API keys
- Keep `.env` in `.gitignore`
- Use environment variables

❌ DON'T:

- Hardcode API keys in code
- Commit `.env` to GitHub
- Share your API key publicly

💡 Pro Tips

1. **Custom Domain**: You can use a custom URL in Streamlit settings
2. **Sleep Mode**: Free apps sleep after inactivity - first visit wakes it up
3. **Resource Limits**: Free tier has limits - optimize if needed
4. **Logs**: Always check logs if something breaks

📞 Support

If you encounter issues:

1. **Check Streamlit Docs**: docs.streamlit.io
2. **Streamlit Forum**: discuss.streamlit.io

3. **Gemini API Docs**: ai.google.dev

✨ You're All Set!

Your User Dashboard is now:

- ✅ Deployed online
- ✅ Accessible via public URL
- ✅ Ready for submission

Next step: Build the Admin Dashboard (Task 2 Part 2)!

Good luck with your assignment! 🚀